



INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

ASSIGNMENT

Assignment Title: Linux System Administration Adventure

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STORYLINE

You have just joined TechVille Inc. as a Junior Linux System Administrator. Your manager has given you a set of tasks to prove your skills. These tasks involve managing users, groups, file permissions, and packages on the company's server.

PART ONE

1. Alice joined the company, you need to create an account for alice using the 'sudo adduser (username)' in this case the username is alice.

```
(cyberfarmer@kali)-[~]
$ sudo adduser alice
[sudo] password for cyberfarmer:
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for alice
Enter the new value, or press ENTER for the default
    Full Name []: Alice
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n]
(cyberfarmer@kali)-[~]
$
```

This command interactively creates a new user named Alice. It prompts for a password and additional details like full name and contact info, and sets up her home directory.

2. Bob is also hired quickly but you do not have time to go through all these prompts so you create an account using 'sudo useradd (username)' which is 'Bob' which would his account non-interactively

```
(cyberfarmer@kali)-[~]
$ sudo useradd bob

(cyberfarmer@kali)-[~]
$
```

Unlike the sudo adduser command, it does not create a home directory or set a password for the user.

3. Bob Resigns from the company as a junior linux administrator you are charged with removing his account and check user list to confirm

```

(cyberfarmer@kali)-[~]
$ sudo deluser bob && getent passwd
root:x:0:0:root:/root:/usr/bin/zsh
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
Debian-gdm:x:129:133:Gnome Display Manager:/var/lib/gdm3:/bin/false
cyberfarmer:x:1000:1000:Garba Ahmed,,,:/home/cyberfarmer:/usr/bin/zsh
systemd-oom:x:983:983:systemd Userspace OOM Killer:/usr/sbin/nologin
pipewire:x:982:125:system user for pipewire:/nonexistent:/usr/sbin/nologin
alice:x:1001:1001:Alice,,,:/home/alice:/bin/bash

(cyberfarmer@kali)-[~]
$

```

The command deleted Bob's account and then listed all users on the system by querying the `/etc/passwd` database.

4. Alice wants to set a more secure password for her account

```

(cyberfarmer@kali)-[~]
$ sudo passwd alice
New password:
Retype new password:
passwd: password updated successfully

```

The command `'sudo passwd alice'` is used to change login password credentials. After the new password is entered, it would ask you to confirm by re-entering the password.

5. Your manager asked you to provide alice's User ID (UID), Group ID (GID), and groups for documentation.

```

(cyberfarmer@kali)-[~]
$ id alice
uid=1001(alice) gid=1001(alice) groups=1001(alice),100(users)

(cyberfarmer@kali)-[~]
$

```

The command `'id alice'` allows you to view those details requested by the manager. These details are important for documentation and access control.

6. You need to test if the account `alice` works properly so you switch to it.

```
(cyberfarmer@kali)-[~]  
$ su alice  
Password:  
(alice@kali)-[/home/cyberfarmer]  
$
```

The command 'su alice' was used to test whether her account is functional and properly configured.

7. HR updates alice's User ID, and you must change it on the system

```
(cyberfarmer@kali)-[~]  
$ sudo usermod -u 1501 alice  
  
(cyberfarmer@kali)-[~]  
$ id alice  
uid=1501(alice) gid=1001(alice) groups=1001(alice),100(users)  
  
(cyberfarmer@kali)-[~]  
$
```

'sudo usermod -u 1501 alice' updates Alice's UID to 1501. This might be needed to match company standards or resolve conflicts.

8. Alice goes on leave so temporarily lock her account

```
(cyberfarmer@kali)-[~]  
$ sudo usermod -L alice  
[sudo] password for cyberfarmer:  
  
(cyberfarmer@kali)-[~]  
$ su alice  
Password:  
su: Authentication failure  
  
(cyberfarmer@kali)-[~]  
$
```

The 'sudo usermod -L alice' would temporarily disable Alice's login by locking her password. She won't be able to log in because she is unauthorized until unlocked, then I tried to switching user to alice and it prompted 'su: Authentication failure.'

9. Alice returns from leave and you unlock her account

```
(cyberfarmer@kali)-[~]  
$ sudo usermod -U alice  
  
(cyberfarmer@kali)-[~]  
$ su alice  
Password:  
(alice@kali)-[/home/cyberfarmer]  
$
```

This command 'sudo usermod -U alicia' would re-enable Alice's login given her access to the account by unlocking her password.

10. Alice changed her name to Alicia, as a junior linux administrator you are tasked with changing her username from alicia to Alicia using the 'sudo usermod -l alicia alicia'.

```
(cyberfarmer@kali)-[~]
$ sudo usermod -l alicia alicia
[sudo] password for cyberfarmer:

(cyberfarmer@kali)-[~]
$ cat /etc/passwd
root:x:0:0:root:/root:/usr/bin/zsh
```

After that, I used the 'cat /etc/passwd' which lists all the users on the Kali Operating System to confirm that the name change was successful.

11. The company updates alicias group ID to 1601. In other to achieve that, the groupname must be changed from alicia to alicia because the previous command 'sudo usermod -l alicia alicia' only changed the username and not the group name so 'sudo groupmod -n alicia alicia' was firstly inputted to change the group name

```
(cyberfarmer@kali)-[~]
$ sudo groupmod -n alicia alicia

(cyberfarmer@kali)-[~]
$ cat /etc/passwd
root:x:0:0:root:/root:/usr/bin/zsh
```

```
(cyberfarmer@kali)-[~]
$ sudo groupmod -g 1601 alicia

(cyberfarmer@kali)-[~]
$ id alicia
uid=1001(alicia) gid=1601(alicia) groups=1601(alicia),100(users)

(cyberfarmer@kali)-[~]
$
```

From the second attached image above, 'sudo groupmod -g 1601 alicia' was inputted to change the group ID (GID) of the user.

12. HR requests you to update Alicia's full name to 'Alicia Johnson'.

```
(cyberfarmer@kali)-[~]
$ sudo usermod -c "Alicia Johnson" alicia
```

This command sets Alicia's full name in the comment field of '/etc/passwd'. Which is very useful for identification.

13. For security reasons, Alicia has to change her password again, we now use the 'sudo passwd alicia' this time and not alicia.

```
(cyberfarmer@kali)-[~]  
$ sudo passwd alicia  
New password:  
Retype new password:  
passwd: password updated successfully  
  
(cyberfarmer@kali)-[~]  
$
```

PART TWO

1. The Company decides to add a department called developers department

```
(cyberfarmer@kali)-[~]  
$ sudo groupadd developers  
[sudo] password for cyberfarmer:  
  
(cyberfarmer@kali)-[~]  
$
```

This command 'sudo groupadd developers' would automatically create a new group 'developers' which is useful for managing permissions collectively.

2. An 'admins' department is created with a special GID (5050).

```
(cyberfarmer@kali)-[~]  
$ sudo groupadd -g 5050 admins  
  
(cyberfarmer@kali)-[~]  
$
```

Unlike the 'sudo groupadd developers' command, this command (sudo groupadd -g 5050 admins) would also create a group admin but with a GID of 5050.

3. The admin group was created by mistake, now you have to delete it.

```
(cyberfarmer@kali)-[~]  
$ sudo groupdel admins  
  
(cyberfarmer@kali)-[~]  
$
```

The command 'sudo groupdel admins' would delete or remove the admins group created by mistake.

4. You need to confirm the list of all groups in the system.

```
(cyberfarmer@kali)-[~]  
$ getent group  
root:x:0:  
daemon:x:1:  
bin:x:2:  
sys:x:3:  
adm:x:4:cyberfarmer  
tty:x:5:
```

The 'getent group' command would provide a listing of all groups in the operating system.

5. Alicia joins the developers team so in order to add her, we use 'sudo usermod -aG developers alicia'

```
(cyberfarmer@kali)-[~]  
$ sudo usermod -aG developers alicia  
  
(cyberfarmer@kali)-[~]  
$
```

This command is used to add a user into a group.

6. Alicia is moved to another department so you remove

```
(cyberfarmer@kali)-[~]  
$ sudo gpasswd -d alicia developers  
[sudo] password for cyberfarmer:  
Removing user alicia from group developers  
  
(cyberfarmer@kali)-[~]  
$
```

The command 'sudo gpasswd -d alicia developers' would delete alicia from the developers group. This is useful when she changes departments.

7. Confirm Alicia's current group memberships for reporting.

```
(cyberfarmer@kali)-[~]  
$ id alicia  
uid=1501(alicia) gid=1601(alicia) groups=1601(alicia),100(users)  
  
(cyberfarmer@kali)-[~]  
$
```

The command 'id alicia' was used to confirm Alicia's UID, GID, and all the groups she belongs to. This command helps in verifying changes.

PART THREE

1. You create a project file and want to check its default permissions.

```
(cyberfarmer@kali)-[~]  
$ touch project.txt; ls -l project.txt  
-rw-rw-r-- 1 cyberfarmer cyberfarmer 0 Oct 14 15:39 project.txt
```

The 'touch project.txt; ls -l project.txt' would not only creates an empty file but also lists its permissions.

2. Set permissions so that the owner has full access, and others have read and execute.

```
(cyberfarmer@kali)-[~]  
$ chmod 755 project.txt  
  
(cyberfarmer@kali)-[~]  
$ ls -l project.txt  
-rwxr-xr-x 1 cyberfarmer cyberfarmer 0 Oct 14 15:39 project.txt
```

'chmod 755 project.txt' Sets permissions to `rwxr-xr-x`. Owner can read/write/execute; group can read and execute likewise others others can read and execute.

3. Give only the owner (user) execute permission using symbolic notation.

```
(cyberfarmer@kali)-[~]  
$ chmod u+x project.txt  
  
(cyberfarmer@kali)-[~]  
$ ls -l project.txt  
-rwxr-xr-x 1 cyberfarmer cyberfarmer 0 Oct 14 15:39 project.txt  
  
(cyberfarmer@kali)-[~]  
$
```

The 'chmod u+x project.txt' command adds execute permission for the owner only, without affecting others.

4. Assign the developers group ownership of the project file.

```
(cyberfarmer@kali)-[~]  
$ sudo chgrp developers project.txt  
[sudo] password for cyberfarmer:  
  
(cyberfarmer@kali)-[~]  
$
```


The 'sudo chgrp developers project.txt' changes the group ownership to developers. Group members can now access based on group permissions.

5. Make Alicia the owner of the project file.

```
(cyberfarmer@kali)-[~]  
$ sudo chown alicia project.txt  
  
(cyberfarmer@kali)-[~]  
$
```

The 'sudo chown alicia project.txt' transfers file ownership to Alicia.

6. Change both the owner and group to Alicia:developers.

```
(cyberfarmer@kali)-[~]  
$ sudo chown alicia:developers project.txt  
  
(cyberfarmer@kali)-[~]  
$
```

The command would change both owner and group in one command.

7. Verify the permissions and ownership after each step

```
(cyberfarmer@kali)-[~]  
$ ls -l project.txt  
-rwxr-xr-x 1 alicia developers 0 Oct 14 15:39 project.txt  
  
(cyberfarmer@kali)-[~]  
$
```

'ls -l project.txt' displays updated permissions, owner, and group for the project.txt file.

8. Try entering a restricted directory like /root with a normal user. Record what you observed.

```
(cyberfarmer@kali)-[~]  
$ cd /root  
cd: permission denied: /root
```

The error 'cd /root' command didn't work because the /root directory is the reserved home directory for the system's **superuser** (root), and Linux security permissions prevent standard, non-root users are blocked from accessing its contents. By default, the /root directory is secured with restrictive permissions that only the owner (root) has the necessary execute (access) and read permissions to navigate into it.

PART FOUR

1. The developers need wget installed for downloading files.

```
(cyberfarmer@kali)-[~]  
$ sudo apt install wget -y  
[sudo] password for cyberfarmer:  
wget is already the newest version (1.25.0-2).  
The following packages were automatically installed and are no longer r
```

'sudo apt install wget -y' would Install the wget app which is a tool for downloading files from the web. '-y' auto-confirms prompts.

2. Later, wget is no longer needed, so remove it.

```
(cyberfarmer@kali)-[~]  
$ sudo apt remove wget -y  
The following packages were automatically installed and are no longer r  
equired:  
  amass-common  
  binutils-mingw-w64-i686
```

This command would delete the wget app.

3. Update the package list to fetch the latest software information.

we would now use the 'sudo apt update' command

```
(cyberfarmer@kali)-[~]  
$ sudo apt update  
Hit:1 http://http.kali.org/kali kali-rolling InRelease  
41 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

4. Upgrade all installed packages to their latest versions.

```
(cyberfarmer@kali)-[~]  
$ sudo apt upgrade -y  
The following packages were automatically installed and are no longer r  
equired:  
  amass-common  
  binutils-mingw-w64-i686
```

The 'sudo apt upgrade -y' Installs the latest versions of all installed packages.

5. A developer requests curl, so you search for it in the repositories.

```
(cyberfarmer@kali)-[~]  
$ apt search curl  
ario/kali-rolling 1.6-1.5 amd64  
  GTK+ client for the Music Player Daemon (MPD)  
  
ario-common/kali-rolling 1.6-1.5 all  
  GTK+ client for the Music Player Daemon (MPD) (Common files)  
  
cht.sh/kali-rolling 0.0~git20220418.571377f-2 all
```

This 'apt search curl' would search for curl and related packages in the repository which provides useful information before installation.